

EBB Source Roadmap

A single catalog of every data source EBB touches: live and ingesting, configured but disabled, planned but not yet built, or blocked on something outside our control. `ROADMAP.md` tracks system *capabilities* (phases, features); this tracks *sources* specifically — one place to answer “do we have X yet, and if not, why not.”

For day-to-day source problems (expired tokens, 403s, stale cursors), see the generated `HEALTH_ACTIONS.md` (`python3 health_report.py`). For manual remediation steps (registrations, backfills), see `MAINTENANCE.md`. This document is hand-curated and only needs updating when a source is added, retired, or its plan status changes — not on every run.

Snapshot: 76 sources configured, 71 enabled, 7 requiring auth.

Active — Greenhouse Gases (26 sources)

Mauna Loa (MLO), Barrow/Utqiagvik (BRW), Cape Grim (CGO), South Pole (SPO), American Samoa (SMO), Alert (ALT), Summit (SUM). CO /CH via NOAA CCGG in-situ/flask; N O/SF via CATS gas chromatograph (daily, FTP) or global monthly means.

Species	Stations	Cadence	Notes
CO	MLO (trends + CCGG), BRW, CGO, SPO, SMO, ALT, SUM	daily	Both <code>noaa_co2_daily_mlo</code> (trends, most-current tail) and <code>noaa_co2_daily_mlo_ccgg</code> (full history) are active and complementary
CH	MLO, BRW, CGO, SPO, SMO, ALT, SUM	daily	<code>noaa_ch4_daily_mlo</code> (trends variant) is disabled — superseded by <code>noaa_ch4_daily_mlo_ccgg</code> (fuller history); the CO equivalent above was kept enabled instead

Species	Stations	Cadence	Notes
N O	MLO, BRW, SPO, SMO, SUM (CATS, FTP) + global monthly mean	daily / monthly	CATS instruments at MLO/BRW/SMO/SPO/SUM were decommissioned ~2022 — these five sources are permanently frozen at their last CATS reading, not a fetch bug (see <code>HEALTH_ACTIONS.md</code>)
SF	MLO, BRW, SPO, SMO, SUM (CATS, FTP) + global monthly mean	daily / monthly	same CATS end-of-life caveat as N O

Known systemic issue: CCGG in-situ/flask data (CO₂, CH₄ across all stations) is currently ~230 days behind real-time upstream at NOAA GML — confirmed by direct fetch, not an EBB bug. QC/lab-processing lag, not a fetch problem.

Active — Ocean (23 sources)

Indicator	Source	Cadence	Notes
SST — 8 stations	NOAA OISST v2.1 ERDDAP (equatorial Pacific, N. Atlantic, Southern Ocean, Arctic Beaufort, Coral Sea, 3× Subpolar Gyre)	daily	SPG stations are the AMOC proxy (Caesar et al.)
Global mean sea level	AVISO merged NRT altimetry (<code>aviso_gmsl_merged_nrt</code>)	weekly	replaced frozen <code>noaa_gmsl_satellite</code> (disabled)

Indicator	Source	Cadence	Notes
Ocean heat content 0–700m	NOAA WOA22 (4 quarterly source entries)	quarterly	
Ocean heat content 0–2000m	NOAA WOA22 (4 quarterly source entries)	quarterly	full Argo-depth extension of the above (§21)
ENSO — Niño 3.4	NOAA CPC <code>sstoi.indices</code>	monthly	
ENSO — ONI	NOAA CPC <code>oni.ascii.txt</code>	monthly	
PDO	NOAA PSL	monthly	
AMO	NOAA PSL	annual	
SST → global mean (ERA5)	Copernicus CDS	monthly	requires CDS key
Coral bleaching (DHW)	NOAA Coral Reef Watch ERDDAP	daily	global max degree-heating-weeks

Active — Atmosphere & Circulation Indices (3 sources)

Indicator	Source	Cadence
NAO	NOAA CPC	monthly
TCWV (water vapor)	ERA5 / Copernicus CDS	monthly
Global/land temp anomaly (ERA5 cross-check)	Copernicus CDS	monthly

Active — Land, Cryosphere & Biosphere (8 sources)

Indicator	Source	Cadence	Notes
Global surface temp anomaly (land+ocean, land-only)	NOAA NCEI Climate at a Glance	monthly	
US Palmer Drought Severity Index	NOAA CPC/NCEI	monthly	supersedes blocked NDMC
Arctic sea ice extent (N + S)	NSIDC	daily	

Indicator	Source	Cadence	Notes
Arctic sea ice volume	PIOMAS v2.1 (Univ. Washington)	daily	URL has a {year} template that needs the annual filename rollover watched
Antarctic ozone hole	NASA Ozone Watch	annual	
NH snow cover extent	Rutgers Global Snow Lab	monthly	ice-albedo feedback proxy (§2j)

Active — Events (7 sources)

Source	Coverage	Cadence
USGS	Earthquakes M4.5+, global	daily
EMSC (seismicportal.eu)	Earthquakes M3.5+, Europe/ME/Central Asia	daily
NASA EONET	Multi-hazard (fire, storm, volcano, flood)	daily
GDACS	Multi-hazard, Orange/Red alerts	daily
NASA FIRMS (VIIRS SNPP)	Global active-fire thermal anomalies	daily
WMO SWIC	Active tropical cyclone advisories, all basins	daily
IBTrACS (ibtracs_last3years)	Historical tropical cyclone best-track	weekly

Cross-source duplicates (same earthquake/fire/storm reported by multiple feeds) are suppressed viewer-side by DEDUP_RULES — see ROADMAP §2i.

Active — Ice Sheet Mass (auth-gated, 2 sources)

Indicator	Source	Cadence	Auth
Greenland mass anomaly	GRACE-FO RL06.3 (JPL)	monthly	Earthdata token
Antarctica mass anomaly	GRACE-FO RL06.3 (JPL)	monthly	Earthdata token

Active — Conflict (2 sources, both degraded)

Source	Status	Detail
ACLED	auth rejected	OAuth login succeeds; data API returns 403 — account needs API access enabled on the ACLED dashboard. See MAINTENANCE.md .
UCDP	broken	URL points at the API docs page, not a data endpoint — parser gets HTML, extracts 0 rows every run.

Disabled — templates / superseded / manual-only (5 sources)

Source	Why disabled
noaa_ch4_daily_mlo	Superseded by <code>noaa_ch4_daily_mlo_ccgg</code> (same data, fuller history)
noaa_gmsl_satellite	Superseded by <code>aviso_gmsl_merged_nrt</code> after the NOAA LSA feed went stale
rapid_amoc	Real dataset, real parser — needs manual registration + one-off download; see MAINTENANCE.md . Not auto-fetchable, so left disabled rather than erroring daily
gov_official_releases_example / parliament_feed_example	Config templates for the political-source category — no real feed wired up yet

Planned — not yet implemented

Grouped by what's blocking them. “Ready” means no known blocker, just not built yet; effort is roughly source-count + parser-count.

Ready to build (no auth/API blocker known)

Indicator	Source	Notes
NHC active advisories (Atlantic/E. Pacific RSS)	<code>nhc.noaa.gov</code> RSS/XML	Complements WMO SWIC with basin-level advisory detail
Annual ACE / named-storm-count snapshot	Derived from IBTrACS archive	Needs <code>ibtracs.since1980</code> one-off backfill first (see <code>MAINTENANCE.md</code>)
Global wildfire burned area (MCD64A1)	NASA LPDAAC	Earthdata token already held; monthly 500m aggregate, complements FIRMS/EONET fire <i>events</i> with an area metric
Deforestation / tree cover loss	Hansen/UMD via Global Forest Watch API	Needs a GFW API key
WDCGG (non-US GHG stations)	<code>gaw.kishou.go.jp</code>	Adds European/Asian/Southern-Hemisphere stations NOAA GML doesn't cover; format/licensing not yet investigated
CAMS global fire radiative power	Copernicus ADS	Have key; daily, complements FIRMS
C3S climate indicator bulletins	Copernicus	Pre-computed monthly sea ice/sea level/glacier JSON/CSV
UN GA voting records	UN	Treaty ratification / resolution voting
FAO food price index	FAO	Monthly commodity prices
UNHCR displacement	UNHCR	Refugee/IDP counts
WHO outbreak reports	WHO	Disease event feed

Needs NASA Earthdata token (already held — just unimplemented)

Indicator	Product	Notes
Soil moisture	SMAP L3 (SPL3SMP), NSIDC DAAC	Daily 36km global; key drought/land-stress signal currently missing

Indicator	Product	Notes
Arctic surface albedo	MODIS MCD43C3	Ice-albedo feedback (§2j)
NDVI vegetation greenness	MODIS MOD13A3	Amazon/forest dieback feedback (§2j)
Atmospheric XCO	OCO-2/OCO-3 L2 lite files, GESDISC	~16-day revisit; validates surface in-situ
Total ozone column (daily, not just annual hole)	TROPOMI/OMI L3, GESDISC/Copernicus mirror	Complements NASA Ozone Watch's annual hole metric
Sea ice thickness	ICESat-2 ATL20, NSIDC DAAC	Complements PIOMAS volume + NSIDC extent
Global temp cross-check	GISTEMP v4 (GISS)	Different methodology from NOAA NCEI — cross-validation value

Needs Copernicus ADS/CDS (have key — just unimplemented)

Indicator	Product	Notes
Atmospheric methane column (XCH4)	TROPOMI/Sentinel-5P	Highest-value unmonitored permafrost-feedback signal — localizes Arctic methane hotspots that surface stations can't
Soil temperature	ERA5 st11/st12	Permafrost feedback
Aerosol optical depth	CAMS	Aerosol-masking-removal feedback — IMO 2020 shipping rules may be unmasking warming
SO2 column	TROPOMI	Same feedback theme as AOD

Needs a new credential/registration

Indicator	Source	Blocker
SPEI global drought index	spei.csic.es	Free HTTP, just not investigated/wired up yet — lowest-friction item on this list

Indicator	Source	Blocker
Permafrost extent / ALT / ground temp	ESA CCI Permafrost v5.0, THREDDS	No auth, but 12–360MB NetCDF files need OPeNDAP subsetting for the largest product; supersedes GTN-P (no usable API)
AMOC reanalysis ensemble	CMEMS GLOBAL_OMI_NATLANTIC_amoc	Needs an open 26N time series package + a <i>separate</i> Copernicus Marine account (distinct from CDS)

Blocked — no viable path currently

Indicator	Source	Why blocked
AMOC transport at 26°N (direct observation)	RAPID MOCHA-WBTS	Highest scientific value of anything on this list — the only direct long-term AMOC record. Requires emailing <code>rapid@noc.ac.uk</code> for registration, then a manual annual download. Parser (<code>parse_rapid_amoc_netcdf</code>) and source config already exist, disabled, waiting on registration.
Glacier mass balance	WGMS	Data-order form only; no open API; all direct HTTP returns 403/404
US drought statistics	NDMC	Documented API patterns all 404; server-rendered ASP.NET. Superseded by NOAA PDSI, which is live.

Indicator	Source	Why blocked
Humanitarian event feed	ReliefWeb	v1 API gone (410), v2 forbidden (403) — likely deprecated/restricted
Indian Ocean Dipole (DMI)	—	No reliable free HTTP endpoint found as of 2026-08: JAMSTEC access suspended, NOAA PSL URL 404s, BOM 403s

Superseded

Old	New	Why
GTN-P (permafrost)	ESA CCI Permafrost v5.0 (planned, see above)	GTN-P is an Angular 1.x app with GeoServer WMS only — no data API
NOAA LSA satellite altimetry (GMSL)	AVISO merged NRT	NOAA feed went stale/frozen

Maintaining this document

Update when: a planned source ships (move it to Active with its config entry), a blocked source gets unblocked (registration approved, new API found), or a new candidate source is identified. Don't update this for day-to-day health fluctuations — that's what `health_report.py` is for.